

What are magnets?

Scientists think that atoms behave like tiny magnets. In a nonmagnetic material, the atoms face in different directions. In a magnet, all the atoms face the same way. Magnets lose their strength if they are hit, dropped, or heated, as this makes the atoms face in different directions.

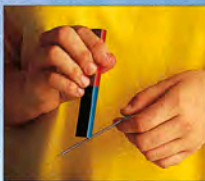
The ends of a magnet are called its poles.



It is hard to push two identical poles together. The magnetic forces are pushing them apart.

Every magnet has two poles, a south pole (S), and a north pole (N).

It isn't easy to pull two different poles apart. The magnetic forces between them hold them close together.

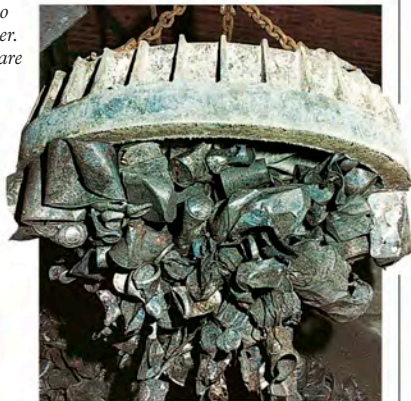
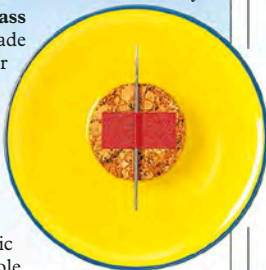


Make a magnet

If you stroke a needle about 50 times with a magnet, it will become magnetic. Stroking the needle in the same direction turns its atoms to face the same way.

Simple compass

Compasses are made with magnets—see for yourself! Tape your magnetic needle to a piece of cork, and float the cork in some water. It points the same way as a real compass needle—toward the magnetic North Pole.



Electricity and magnets

Electricity is used to make magnets which can be switched on and off. The amount of electric current determines the strength of the magnetic field produced. Reversing the flow of electricity will also reverse the magnet's poles. This electromagnet is used to sort scrap metal.